

Muhammad Hassan

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EDUCATION

Virginia Tech

PhD in Computer Science - CGPA **4.0/4.0**

Jan 2025 - Dec 2029 (expected)

Blacksburg, Virginia

Lahore University of Management Sciences

Bachelor of Science in Computer Science – Major GPA **3.70/4.00**

Sep 2019 - Jun 2023

Lahore, Pakistan

PUBLICATIONS

- ***Taming the Hydra: Targeted Control-Flow Transformations for Dynamic Symbolic Execution.*** C. Saumya*, **M. Hassan***, et al. PACMPL (OOPSLA), 2026.
- ***[ThunderFork: Splitting SMT Queries with Targeted Control-Flow Transformations.](#)*** **M. Hassan.** SPLASH/ISSTA Student Research Competition, Extended Abstract, 2026.
- ***Evaluating Container Debloaters.*** **M. Hassan***, T. Tahir*, et al. IEEE SecDev, 2023.

HONORS AND AWARDS

- **Pratt Fellowship**, Virginia Tech **May 2026**
- **OOPSLA Artifact Badges** (Functional, Reusable, Reproduced) for Taming the Hydra **Mar 2026**

EXPERIENCE

Virginia Tech - Language And Compiler Design Lab

Graduate Research Assistant

Jan 2025 - Present

Blacksburg, Virginia

- Developed **LLVM-based control-flow transformations** for KLEE, reducing symbolic execution time from 20 hours to 4 hours on selected benchmarks.
- Developed **solver-agnostic, cost-guided control-flow unmerging** for parallel SMT solving, achieving **25 wins, 18 ties, and 1 loss against native Z3**, and **17 wins, 9 ties, and 18 mutual timeouts** against native cvc5 across 44 benchmarks.
- Developing **constant-time compiler transformations** to mitigate timing side channels.

Aerodyne Group

Software Engineer

Jul 2023 - Dec 2024

Lahore, Pakistan

- Developed a 2D/3D annotation interface for digital-twin models.
- Optimized the 3D rendering pipeline, increasing frame rate from 15 FPS to 240 FPS.

Lahore University of Management Sciences

Research Assistant

Jan 2023 - Oct 2024

Lahore, Pakistan

- Developed and evaluated BloatProfiler, a benchmarking framework for Docker image debloating.
- Analyzed CVE mitigation, correctness, and size reduction of debloated images.

PROJECTS

LLM-Based Debloating: Developed a pipeline that complements debloating techniques using Azure AI GPT-3.5, RAG architecture, and LLVM coverage analysis, resulting in improved generality and security of debloated applications.

SKILLS

Languages: C, C++, Python, Haskell, Assembly, JavaScript/TypeScript

Compilers & Program Analysis: LLVM, Clang, KLEE, Z3, cvc5, STP, SMT-LIB, Static Analysis

Development Tools: Linux, Git, Docker, CMake, Bash